The Short, the Long, and the Consequences of VAR Analysis For Structural DSGE Estimation

Master's Degree in Applied Economics; Economic Data Analytics

University of Padova; Supervisor Prof. Efrem Castelnuovo

Lorenzo Del Col

June 22, 2026

Abstract

Truncation bias could be a severe problem regarding structural VAR analysis (De Greave and Westermark) (DGW2025) claims that the role of lag choice in SVAR is crucial to correctly identify IRFs, and they claim that choosing long lags solve the bias and, surprisingly, this reduce the variance of the point estimate. CEE2005 is a classic paper that studies the effects of monetary policy shock in the economy. Particularly their DSGE is calibrated ex-ante using matching the model's IRFs with the ones estimated from the VAR analysis. The aim of this thesis is to perform the (DGW2025) exercise, see if and how the IRFs changes, analys how the DSGE model calibration vary and if the model is consistent to this new empirical way to estimate the VARs. If so, we want to understand more about the claimed variance reduction caused by the solution of the truncation bias how works in the Frequentist and Bayesian world.

Contents

1 Introduction

Provide the background of your topic, state the research problem, and outline the structure of your thesis here. Maybe I have switched the latex writing in VS Code

1.1 Research Question

Clearly define what your thesis aims to solve or investigate.

2 Theoretical Framework

Discuss the existing literature and the theoretical models that underpin your research.

3 Methodology

Explain the empirical strategy, models, and data you are using.

4 Empirical Results

Present your findings, regression tables, and graphs here.

5 Conclusion

Summarize the main takeaways, discuss the policy implications, and mention any limitations of your work.

References

A Replication Files and Code

Detail where the reader can find your code (e.g., a GitHub link).

B Additional Figures and Tables

B.1 ACEL 2005 VAR Variables

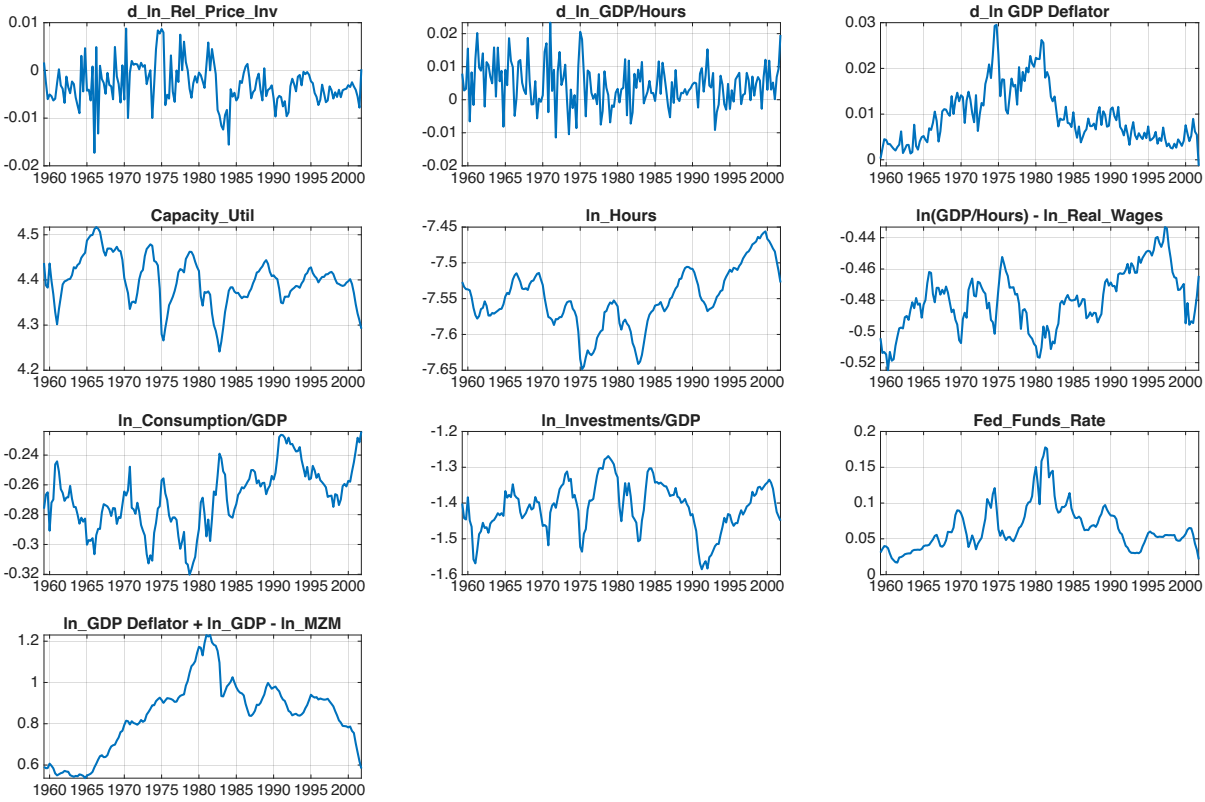


Figure 1: Raw Historical Series

B.2 ACEL 2005 VAR(4) Baseline Shocks Results

B.3 ACEL 2005 VAR(4) DSGE MatchedBaseline Shocks Results

Embodied Technology Shock

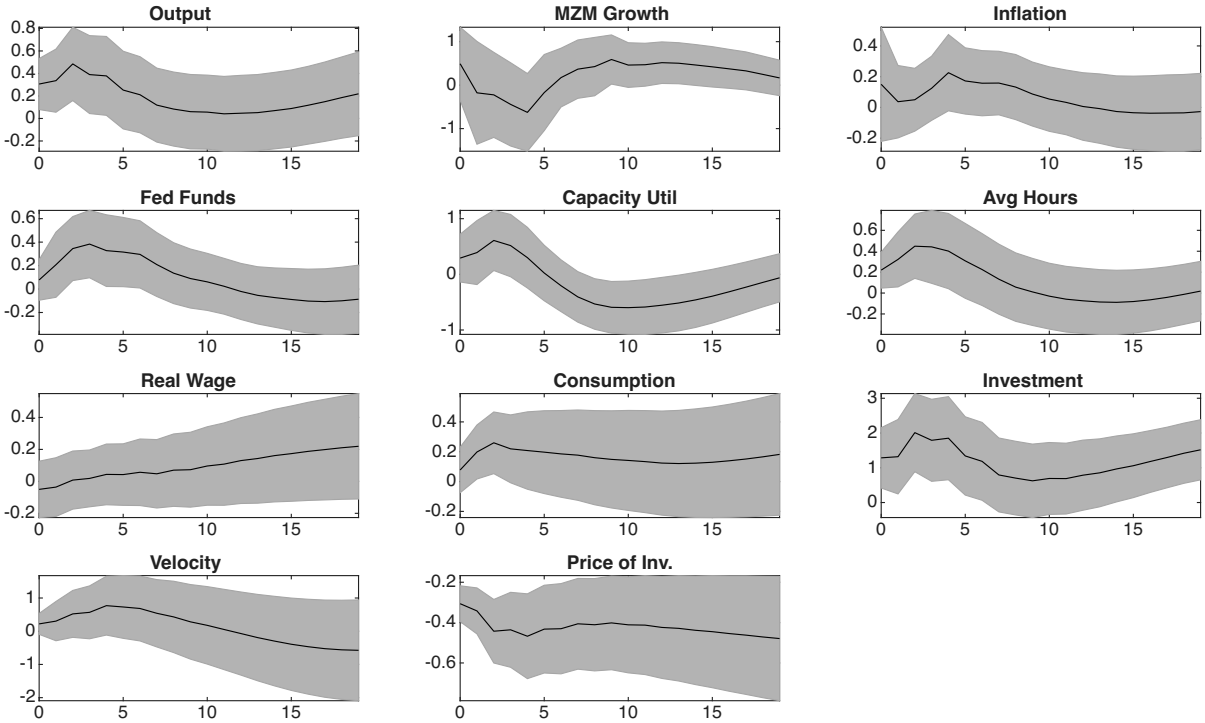


Figure 2: Embodied Technology Shock

Neutral Technology Shock

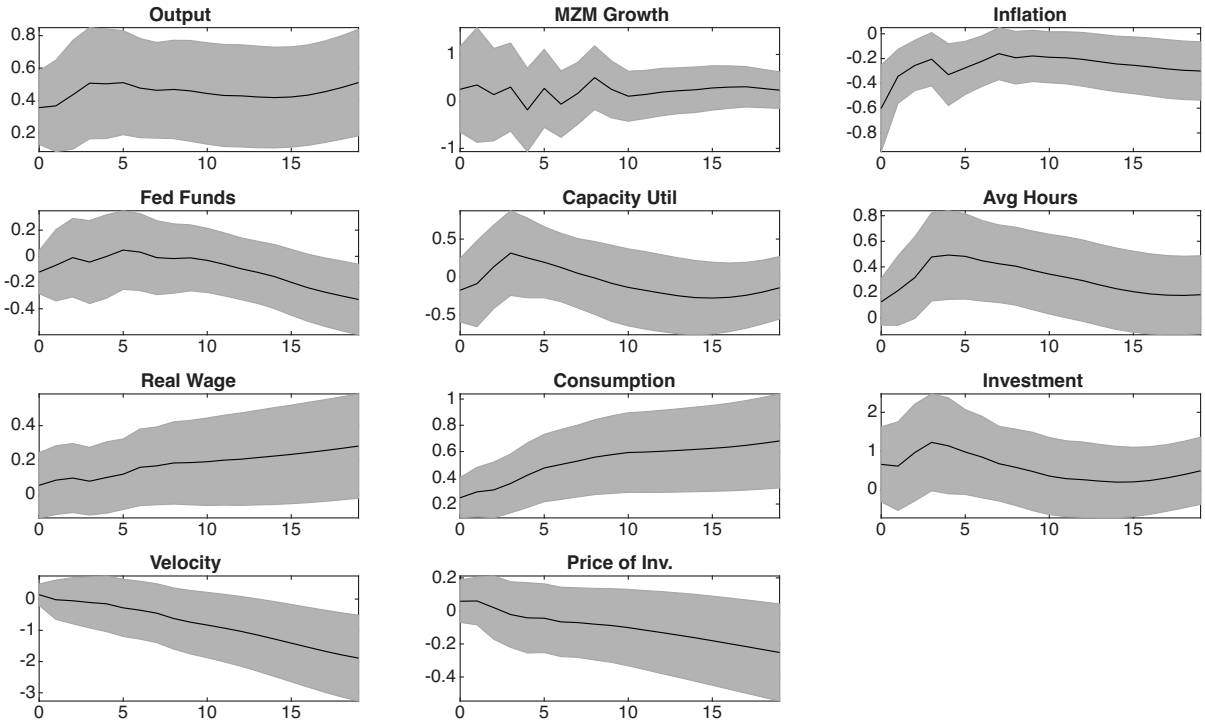


Figure 3: Neutral Technology Shock

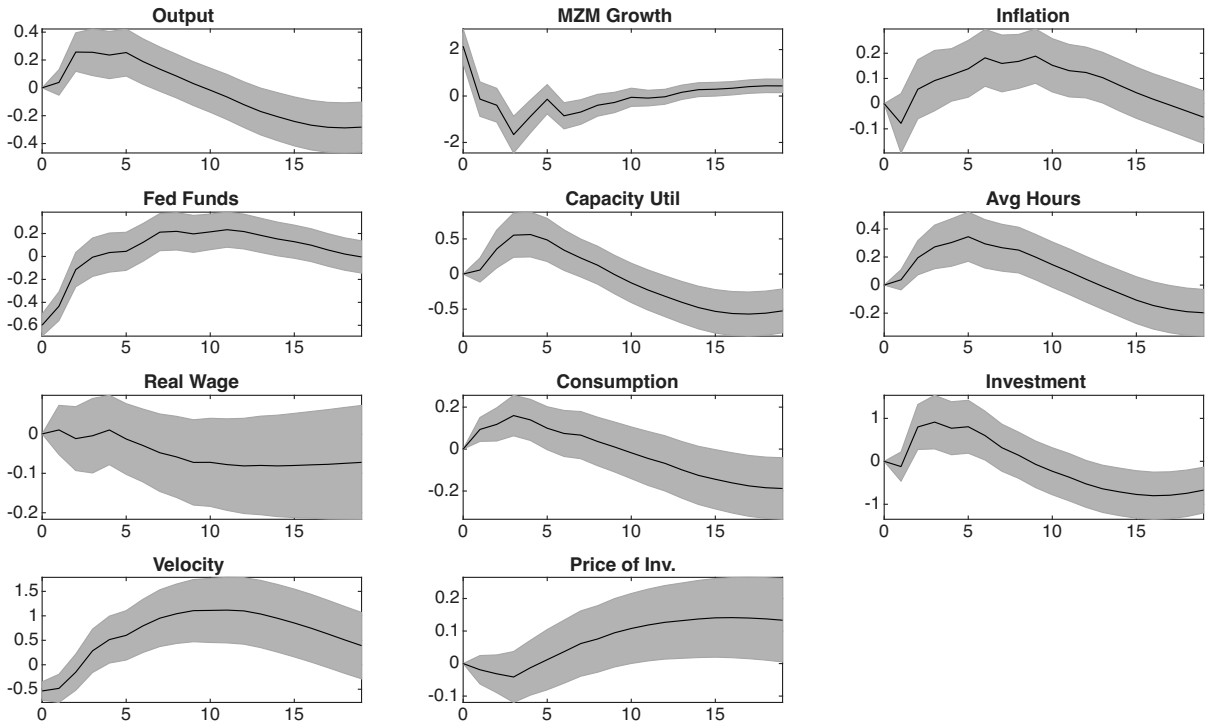


Figure 4: Monetary Policy Shock

Missing file: ../../Output/LDC$_A$CEL2011$_I$mpulse$_R$esponse$_M$atching/ET$_b$asic$_D$SGE/ET$_b$asic$_D$SGE$_l$ag$_4$.pdf

Figure 5: Embodied Technology Shock DSGE4 Matched

Missing file: ../../Output/LDC$_A$CEL2011$_I$mpulse$_R$esponse$_M$atching/NT$_b$asic$_D$SGE/NT$_b$asic$_D$SGE$_l$ag$_4$.pdf

Figure 6: Neutral Technology Shock DSGE4 Matched

Missing file: ../../Output/LDC$_A$CEL2011$_I$mpulse$_R$esponse$_M$atching/MP$_b$asic$_D$SGE/MP$_b$asic$_D$SGE$_l$ag$_4$.pdf

Figure 7: Monetary Policy Shock